

HAT-P-36b

R-0768

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_251208 · created 2026-07-28T04:56:11+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2461017.9157 +/- 0.006 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.112 +/- 0.037
Transit depth (Rp/Rs)^2	1.26 +/- 0.82 [%]
Orbital Inclination (inc)	81.4 +/- 2.77
Airmass coefficient 1 (a1)	1.215 +/- 0.015
Airmass coefficient 2 (a2)	-0.0916 +/- 0.0086
Scatter in the residuals of the lightcurve fit is	1.23 %
Best Comparison Star	None
Optimal Aperture	3.84
Optimal Annulus	14.88
Transit Duration (day)	0.072 +/- 0.018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.112 $\pm$ 0.037 vs 0.126 (deviation 0.38 σ)
Duration fit vs published	0.072 d vs 0.0925 d (deviation 1.14 σ)
Mid-time O-C	-10.1 min
Depth SNR	3.0
Residual scatter	1.23 %

Light curve

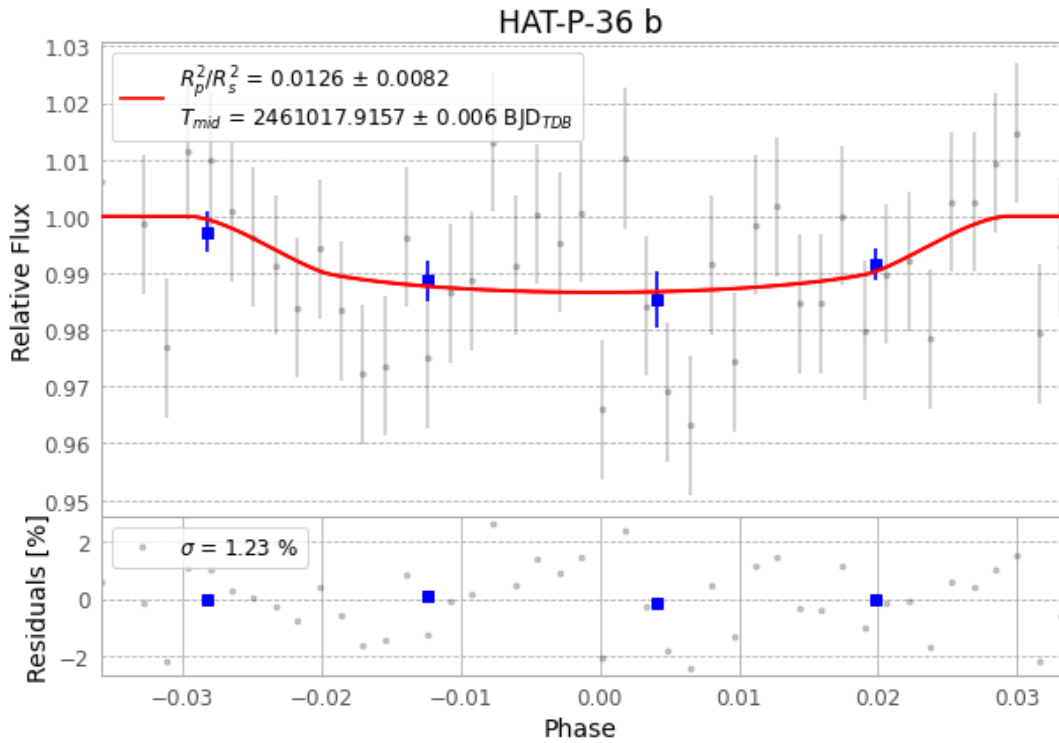


Figure 1 — FinalLightCurve_HAT-P-36 b_2025-12-08.png (SHA-256 6c91ab85117987e8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [434, 181], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 53087790a574dffa...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

44 FITS frames (29 MB), HATP-36251208084816.FITS → HATP-36251208110015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-251208.log (ee48bbff7baa...), FinalLightCurve_HAT-P-36 b_2025-12-08.png (6c91ab851179...), FinalLightCurve_HAT-P-36 b_2025-12-08.csv (97a1732c8f33...)

Compute resources

Wall time 100.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 04:56Z.